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# Interview Questions & Answers



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# What is Context Window in an LLM?

**Context window is the amount of text an LLM can remember and process at one time.**

## It includes:

- User prompt
- Chat history
- Documents
- Instructions

**Measured in tokens.**

## Real-world Example:

**Customer support chatbot:**

- Remembers earlier conversation
- Uses previous messages for better replies

**Without enough context window:**

- Model forgets earlier details

## Why Important?

**Large context windows help in:**

- Long conversations
- Document analysis
- Coding assistants
- AI agents
- Enterprise search

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# What is Quantization in LLMs?

**Quantization reduces model size by lowering numerical precision.**

## Example:

- **FP32 → INT8**
- **FP16 → 4-bit**

## This makes models:

- **Faster**
- **Smaller**
- **Cheaper to run**

## Popular Tools:

- **NVIDIA TensorRT**
- **GGUF**
- **bitsandbytes**

## Benefits:

✅ **Lower memory usage**
✅ **Faster inference**
✅ **Reduced cost**
✅ **Edge deployment possible**

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# What is a Vector Database?

✅Vector databases store embeddings for semantic search.
✅Instead of keyword matching, they search based on meaning.

## How It Works:

1. Convert text into embeddings
2. Store vectors
3. Compare similarity during search

## Common Use Cases:

- RAG systems
- Recommendation engines
- AI search
- Chatbots

## Popular Vector Databases:

- Pinecone
- Weaviate
- Chroma
- FAISS

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# What is Function Calling / Tool Calling in LLMs?

**Function calling allows LLMs to interact with external tools and APIs.**

## The LLM decides:

- **Which tool to use**
- **What parameters to send**

## Real-world Tools:

- **Weather APIs**
- **Database queries**
- **Email sending**
- **Calendar scheduling**
- **Payment systems**

## Popular Frameworks:

- **OpenAI Function Calling**
- **LangChain Tools**
- **CrewAI**
- **AutoGen**

## Why Important?
## Function calling powers:

- **AI agents**
- **Automation systems**
- **Enterprise copilots**
- **Smart assistants**

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